

Unraveling the Intricate Network of lncRNAs in Corneal Epithelial Wound Healing: Insights Into the Regulatory Role of linc17500

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Purpose: Epigenetic mechanisms orchestrate a harmonious process of corneal epithelial wound healing (CEWH). However, the precise role of long non-coding RNAs (lncRNAs) as key epigenetic regulators in mediating CEWH remains elusive. Here, we aimed to elucidate the functional contribution of lncRNAs in regulating CEWH.

Methods: We used a microarray to characterize lncRNA expression profiling during mouse CEWH. Subsequently, the aberrant lncRNAs and their *cis*-associated genes were subjected to comprehensive Gene Ontology (GO) enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses. Quantitative reverse transcription–polymerase chain reaction (RT-qPCR) and western blot analyses were performed to determine the expression profiles of key markers during CEWH. The *in vivo* effects of linc17500 on this process were investigated through targeted small interfering RNA (siRNA) injection. Post-siRNA treatment, corneal re-epithelialization was assessed, alongside the expression of cytokeratins 12 and 14 (Krt12 and Krt14) and Ki67. Effects of linc17500 on mouse corneal epithelial cell (TKE2) proliferation, cell cycle, and migration were assessed by multicellular tumor spheroids (MTS), 5-ethynyl-2'-deoxyuridine (EdU), flow cytometry, and scratch-wound assay, respectively.

Results: Microarray analysis revealed dysregulation of numerous lncRNA candidates during CEWH. Bioinformatic analysis provided valuable annotations regarding the *cis*-associated genes of these lncRNAs. *In vivo* experiments demonstrated that knockdown of linc17500 resulted in delayed CEWH. Furthermore, the knockdown of linc17500 and its *cis*-associated gene, CDC28 protein kinase regulatory subunit 2 (*Cks2*), was found to impede TKE2 cell proliferation and migration. Notably, downregulation of linc17500 in TKE2 cells led to suppression of the activation status of Akt and Rb.

Conclusions: This study sheds light on the significant involvement of lncRNAs in mediating CEWH and highlights the regulatory role of linc17500 on TKE2 cell behavior.

Translational Relevance: These findings provide valuable insights for future therapeutic research aimed at addressing corneal wound complications.

Introduction

The corneal epithelium, positioned as the outermost layer of the cornea, serves as a protective barrier that safeguards against pathogen infiltration into the underlying tissues, due to its structural integrity and tight junctions.¹ Preservation of corneal transparency and visual acuity relies on continuous self-renewal of the corneal epithelium, which is tightly regulated by

a cascade of cytokines governing cell proliferation, migration, and differentiation processes.^{2,3} However, various clinical factors such as trauma, infection, and surgical procedures can induce corneal epithelial injuries, leading to compromised intercellular connections, altered membrane permeability, and impaired barrier function. Failure to achieve efficient wound repair may result in the invasion of external pathogenic factors, ultimately causing tissue edema, impaired corneal transparency, and even severe vision loss.⁴

Despite advancements in our understanding of corneal epithelial wound healing (CEWH), significant knowledge gaps still exist regarding its comprehensive regulatory mechanisms. Notably, recent studies have increasingly highlighted the pivotal role of epigenetic mechanisms in orchestrating a harmonious and coordinated wound healing process.^{5,6}

Epigenetic mechanisms, characterized by their malleability and adaptability, encompass non-coding RNAs, DNA methylation, and histone modification as prominent contributors.⁷ With regard to CEWH, the emerging understanding of epigenetic mechanisms has unveiled the involvement of DNA methylation in promoting the healing process by targeting microRNA-200a (miR-200a) and cyclin-dependent kinase inhibitor 2B (CDKN2B).⁸ Moreover, increased expression of miR-146 and miR-424 has been shown to impede the wound healing process in diabetic corneal epithelium.⁹ Our previous study established miRNA expression profiling of CEWH, identified 29 highly abundant differentially expressed miRNAs, and further revealed that miR-204 and miR-184 may be potential drug targets and biomarkers for CEWH.^{10,11} Additionally, histone modification, such as SUV39H1-H3K9me3, plays a critical role in regulating corneal epithelial cell proliferation during CEWH.¹² Although considerable advancements have been made in unraveling the functions of miRNAs, the comprehensive characterization of long non-coding RNAs (lncRNAs) and their involvement in CEWH remain largely unexplored.

lncRNAs are a class of RNA molecules that are rarely encoded in mammalian cells and are typically larger than 200 nucleotides.¹³ Notably, lncRNAs constitute a substantial proportion of total cellular RNA, accounting for over 90%, surpassing the extensively studied messenger RNAs (mRNAs), miRNAs, and other RNA species.^{14,15} Recent studies have shed light on the involvement of lncRNAs in gene regulation at both transcriptional and translational levels, achieved through interactions with proteins, DNA, and RNA molecules.¹⁶ Additionally, lncRNAs can function as miRNA inducers, exerting regulatory influence on gene expression.¹⁷ Consequently, the discovery of lncRNAs has provided a fresh perspective on genomic regulation, enabling a reevaluation of our understanding of regulatory mechanisms. Extensive investigations into the biological functions of lncRNAs and their regulatory mechanisms in diseases have significantly contributed to a more comprehensive understanding of disease pathogenesis, facilitating the identification of novel diagnostic markers and therapeutic targets. Numerous studies have demonstrated that lncRNAs play pivotal roles as key regulators

in various biological processes, including cell proliferation, migration, invasion, apoptosis, and particularly epithelial-mesenchymal transition (EMT).^{18,19} For example, lncRNA H19 has been implicated in inhibiting p53-mediated intestinal epithelial cell proliferation and regeneration.²⁰ Moreover, exosomal lncRNA H19 released by mesenchymal stem cells (MSCs) has been shown to enhance fibroblast proliferation and migration, thereby expediting the wound healing process in diabetic foot ulcers.²¹ When considering the vital role of cell proliferation, migration, EMT, and tissue remodeling in successful CEWH,³ it is plausible to speculate that lncRNAs play a critical role in regulating the CEWH process based on the available evidence.

In this study, a comprehensive approach combining microarray assays and bioinformatic functional analyses was employed to examine the expression patterns of lncRNAs during CEWH from an epigenetic perspective. Experimental models involving mouse CEWH and TKE2 cells were utilized to investigate the specific contribution of lncRNA NONMMUT017500 (linc17500) in CEWH. The findings from this study will serve as a foundational platform for a deeper comprehension of the role of lncRNAs in the CEWH process and the identification of novel biomarkers for assessing CEWH.

Methods

Murine Model of Corneal Epithelial Wound Healing

All animal treatments were performed in strict accordance with the ARVO Statement for the Use of Animals in Ophthalmic and Vision Research. Male and female C57BL/6 mice, 8 weeks old, were included in equal numbers for the study. Prior to the experimental procedures, mice were anesthetized using intraperitoneal injections of a combination of ketamine (87 mg/kg) and xylazine (13 mg/kg; Sigma-Aldrich, St. Louis, MO). Local anesthesia was also applied to the eyes using proparacaine hydrochloride eye drops (Alcon, Geneva, Switzerland). The full thickness of the corneal epithelium, extending up to the corneal/limbal border, was removed using a 0.5-mm corneal rust ring remover (AlgerBrush II; Alger Equipment Company, Lago Vista, TX). Following the creation of the wound, erythromycin eye ointment was administered to the wounded eyes to prevent bacterial infection. After approximately 48 hours, healing progressed until only 10% of the original wound area remained. Subsequently, the entire corneal epithelium

was collected from both the injured and fellow eyes for RNA isolation.

Microarray Experiment and Data Processing

Total RNAs from the injured and fellow eye corneal epithelium were extracted using TRIzol Reagent (Thermo Fisher Scientific, Waltham, MA). The RNA samples were assessed for their content and integrity using an Agilent 2000 Bioanalyzer (Agilent Technologies, Santa Clara, CA). To profile lncRNAs, miRNAs, and mRNAs, the Affymetrix Mouse Transcriptome Assay 1.0 (MTA 1.0; Applied Bioscience, Santa Clara, CA) was utilized, with 2 µg of total RNA used as input. To minimize individual differences, five corneas were pooled together to constitute a biological repeat. The microarray platform used in this study contained approximately 6 million probes targeting over 23,000 mRNA transcripts and more than 55,000 non-coding transcripts. The microarray was designed based on the mm10 (GRCm38) genome version, with probe design and gene annotation information sourced from RefSeq, Ensembl, and UCSC Known Genes, among others. The RNA samples were amplified and labeled into fluorescent copy DNA (cDNA), followed by hybridization onto the microarrays according to the manufacturer's instructions. The arrays were scanned using the Applied Biosystems GeneChip Scanner 3000 (Thermo Fisher Scientific), and the probe cell intensity data were summarized using the default settings of the GeneChip Command Console Software. The processed data and CEL files have been stored in the National Center for Biotechnology Information Gene Expression Omnibus (GEO). Data can be accessed through <http://www.ncbi.nlm.nih.gov/geo> (accession number: GSE246761). After quantile normalization of the raw data, differentially regulated lncRNAs, miRNAs, and mRNAs between the injured and fellow eye corneal epithelium were identified based on statistical significance using *P* values and false discovery rate filtering. Hierarchical clustering analysis was performed to visualize the expression profiles of different lncRNAs. The microarray experiments were conducted by Bohao Biotechnology (Shanghai, China).

Bioinformatic Functional Analyses of Identifying lncRNA *cis*-Associated Genes

To identify specific lncRNA signatures in the injured and fellow eye corneal epithelium, a threshold was set with fold change (FC) > 2 and ANOVA *P* < 0.05. Differentially expressed lncRNAs were

further analyzed for their regulatory mechanisms, including *cis*-acting and *trans*-acting modes. In the *cis*-acting mode, lncRNAs regulate the expression of protein-coding genes located on the same chromosome. A total of 400 differentially expressed lncRNAs (top 200 upregulated and top 200 downregulated) were selected for *cis*-analysis. The predicted *cis*-associated genes were identified by considering the genes within a 10-kb range extending from both ends of each lncRNA. For functional annotation of the *cis*-associated genes, Gene Ontology (GO) enrichment analysis was performed. The analysis covered biological processes, cellular components, and molecular functions, providing meaningful annotations of the protein-coding genes. The significance of GO categories was determined based on *P* values calculated using Fisher's exact test, with *P* ≤ 0.05 considered significant for identifying enriched GO terms associated with the differentially expressed genes. In addition, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis was conducted to explore pathway clusters associated with the differentially regulated gene profiling. KEGG pathway analysis provides insights into molecular interactions and response networks. The enrichment score, represented as $-\log_{10}(P \text{ values})$, indicates the significance of pathway correlations.

RNA Extraction and Quantitative Reverse Transcription–Polymerase Chain Reaction

Quantitative reverse transcription–polymerase chain reaction (RT-qPCR) was performed to validate the microarray data. Briefly, approximately 1 µg extracted RNAs of injured and fellow eye corneal epithelium was subjected to reverse transcription using a reverse transcription system (Promega Corporation, Madison, WI). The resulting cDNA was then amplified using Power SYBR Green PCR Master Mix (Thermo Fisher Scientific) on an Applied Biosystems 7500 Fast & 7500 Real-Time PCR system (Thermo Fisher Scientific). Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) mRNA level was utilized as the internal normalization control to normalize the gene expression levels.

TKE2 Cell Culture and Lentiviral Infection

TKE2, a murine limbal/corneal epithelium–derived progenitor cell line, was maintained in a defined keratinocyte serum-free medium (DKSFM; Thermo Fisher Scientific) supplemented with 10-ng/mL epidermal growth factor (EGF; Sigma-Aldrich) in a humid-

ified chamber at 37°C with 5% CO₂. After 7 days, cells were treated with dissociation medium (TrypLE Express Enzyme; Thermo Fisher Scientific) for 7 to 10 minutes and then passaged onto 6-, 24-, or 96-well plates. TKE2 cells in the logarithmic growth phase were infected with lentivirus-expressing short-hairpin RNA (shRNA) specifically targeting NONMMUT017500 (sh-linc17500-1/sh-linc17500-2) or Cks2 (sh-Cks2) with lentivirus titers of 3×10^8 transduction units (TU)/mL or 4×10^8 TU/mL. As a negative control, cells were infected with a lentiviral vector carrying a non-targeting shRNA sequence (sh-NC) with lentivirus titers of 5×10^9 TU/mL. The lentiviral vectors were chemically synthesized by GenePharma Co., Ltd. (Shanghai, China).

Cell Proliferation and Flow Cytometric Assay

Cell proliferation assays were performed using the CellTiter 96 AQueous MTS kit (Promega) and Invitrogen Click-iT EdU Cell Proliferation Kit for Imaging, Alexa Fluor 488 dye (C10337; Thermo Fisher Scientific) following the manufacturer's instructions. Briefly, a density of 10,000 cells/well was seeded into a 96-well culture plate after lentiviral infection with sh-linc17500, sh-Cks2, or sh-NC. The cells were incubated for specified time points (24, 48, 72, and 96 hours) and then treated with the multicellular tumor spheroids (MTS) solution reagent. Absorbance was measured at 490 nm using a microtiter plate reader (Molecular Devices, San Jose, CA). For the 5-ethynyl-2'-deoxyuridine (EdU) assay, the fluorescein-stained TKE2 cells after 48 hours of cultivation were imaged with a confocal microscope (Carl Zeiss Meditec, Jena, Germany), followed by the use of ImageJ 1.46r (National Institutes of Health, Bethesda, MD) to quantify the proliferating cells. For cell-cycle analysis, 1×10^6 cells were collected and fixed in chilled 70% ethanol for 12 hours at -20°C. The fixed cells were stained with propidium iodide using the BD Cycletest PLUS DNA Reagent Kit (Becton, Dickinson and Company, Franklin Lakes, NJ) according to the manufacturer's instructions. DNA content was analyzed using a BD FACSCalibur Flow Cytometer.

In Vitro Scratch-Wound Assay

Approximately 5×10^5 mitomycin C-treated, cell-cycle-inactivated cells were seeded into a 12-well culture plate following lentiviral infection with sh-linc17500, sh-Cks2, or sh-NC. After 24 hours, the cell monolayers were subjected to vertical scratch wounds using a 200- μ L pipette tip and applying firm and swift pressure. The cell debris was removed by

washing the monolayers twice with Hanks' Balanced Salt, followed by culture in fresh serum-free medium. An initial photograph was taken immediately (0 hours) using a Leica MZ7.5 microscope (Leica Microsystems, Wetzlar, Germany), and a subsequent photograph was captured 24 hours after the start of culture. The cell-free area was measured and normalized based on the photographs using ImageJ.

Western Blot Analysis

Total protein extracts were collected and processed following standard procedures for western blot (WB) analysis. The expression levels of phosphorylated Akt (p-Akt), total Akt, phosphorylated Rb (p-Rb), total Rb proteins and CKS2 in the cell lysates were quantified using primary antibodies (Cell Signaling Technology, Danvers, MA; Abcam, Cambridge, UK) at a dilution of 1:1000. GAPDH (Cell Signaling Technology) was used as the endogenous control. Densitometry analysis of the WB protein bands was performed using ImageJ software.

In Vivo Wound Healing Assay

Seven C57BL/6 mice, 7 to 8 weeks old, were anesthetized as previously described. The intrastromal injections were performed using a 33-gauge needle with a 30°C bevel attached to a 5- μ L syringe (Hamilton Company, Reno, NV). The needle was inserted into the corneal stroma and advanced 1.5 mm to the corneal limbal area. Subsequently, 1 μ L (100 μ M) linc17500 siRNA, 0.1 μ L of 100-g/mL polyethyleneimine (Polyplus-transfection SA, Illkirch-Graffenstaden, France), and 2 μ L of 10% glucose solution were injected. The fellow eye was injected with 0.1 nmol NC siRNA as a negative control. After a 6-hour incubation period, epithelial abrasion was induced using a 2-mm biopsy punch, and the corneal epithelium within the area was gently removed using a 0.5-mm corneal rust ring remover. The wound area was then stained with sodium fluorescein at 0 and 24 hours after the scraping. The area of the epithelial defect was visualized using a Leica MZ7.5 stereo microscope and measured using ImageJ. The percentage of wound healing rate was calculated by dividing the wound healing range by the initial scraped area.

Histological and Immunofluorescence Staining Procedures

Dissected mouse eyes were embedded in optimal cutting temperature compound (Thermo Fisher

Scientific). Corneal sections of 4- μ m thickness underwent hematoxylin and eosin (H&E) staining to delineate tissue morphology and were subsequently assessed using a stereomicroscope. For immunofluorescence, corneal sections (10 μ m) and TKE2 cells were fixed in 4% paraformaldehyde for 30 minutes, then permeabilized with 0.5% Triton X-100 for 20 minutes and blocked with 5% goat serum in phosphate-buffered saline for 1 hour at room temperature. Primary antibody incubation was performed overnight at 4°C, and slides were then exposed to Invitrogen fluorescence-labeled secondary antibodies for 1 hour at room temperature in a light-protected environment. The sections were washed, mounted with antifade mounting medium with 4',6-diamidino-2-phenylindole (DAPI), and examined by confocal microscopy (Zeiss LSM 880). The antibodies used were as follows: cytokeratin 12 (Krt12, 1:100; Proteintech, Rosemont, IL), cytokeratin 14 (Krt14, 1:100; Sigma-Aldrich), and Ki67 (1:100; Cell Signaling Technology).

Statistical Analysis

The quantitative data from independent triplicate experiments were collected, and statistical analyses between groups were performed using a two-tailed Student's *t*-test. $P < 0.05$ was considered statistically significant ($*P < 0.05$, $**P < 0.01$, and $***P < 0.001$).

Results

lncRNA Expression Profiling and Classification in Mouse Corneal Epithelial Wound Healing

Differential gene and non-coding RNA (ncRNA) expression analysis was performed using Transcriptome Analysis Console (TAC) Software (Thermo Fisher Scientific) to identify changes between the injured and fellow eye corneal epithelium. Out of the 41,655 ncRNAs detected by the microarray, 639 were upregulated and 1006 were downregulated during the process of CEWH (Fig. 1A). These non-coding RNAs were classified into five categories based on their genomic location relative to adjacent protein-encoding genes: sense, antisense, bidirectional, intronic, and intergenic (Fig. 1B). Sixteen differentially expressed lncRNAs were initially identified and further validated by RT-qPCR, considering factors such as large fold change, high abundance, and promoter analysis.²² The results obtained from RT-qPCR were consistent with the microarray data (Fig. 1C), confirming the reliabil-

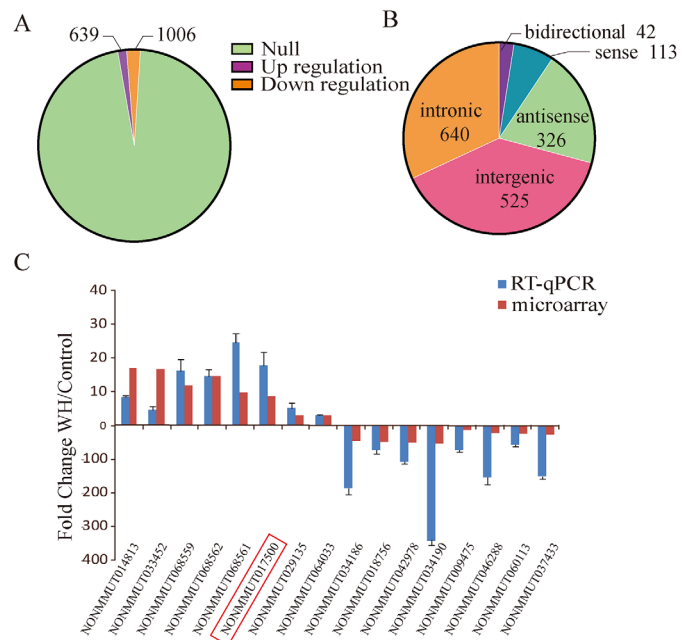


Figure 1. Identification of differentially expressed lncRNA signatures during CEWH through microarray profiling and RT-qPCR. (A) Differentially expressed lncRNAs in repaired corneal epithelia compared with control samples were filtered in terms of FC ≥ 2 or FC < 0.5 . (B) Distribution of the four different catalogs based on their genetic loci corresponding to the filtered lncRNAs. (C) Sixteen highly abundant lncRNAs were selected for validation by RT-qPCR during the wound healing response ($n = 3$ /group). Dysregulated lncRNAs, as determined by RT-qPCR or microarray analysis, are represented by blue or pink columns, respectively.

ity of our microarray analysis. Among these lncRNAs, linc17500 exhibited a significant fold change (FC = 8.53 in the microarray, FC = 17.83 in RT-qPCR) and high expression levels ($\log_2$ signal value = 10.47 in the microarray, cycle threshold = 16.63 in RT-qPCR), making it a suitable candidate for further investigation.

Enrichment Analyses of Differentially Expressed lncRNA *cis*-Associated Genes

Enrichment analyses were conducted to explore the *cis*-associated genes of differentially expressed lncRNAs. A total of 400 lncRNAs (top 200 upregulated and top 200 downregulated) were utilized for the *cis* analysis, which identified the genes located within a 10-kb range upstream and downstream of each lncRNA as predicted *cis*-associated genes. The genomic features of these protein-coding genes in close proximity to the lncRNAs were mapped to the GO database. Significant enrichment of GO terms for differentially expressed genes was determined using a significance threshold of $P \leq 0.05$. The enriched

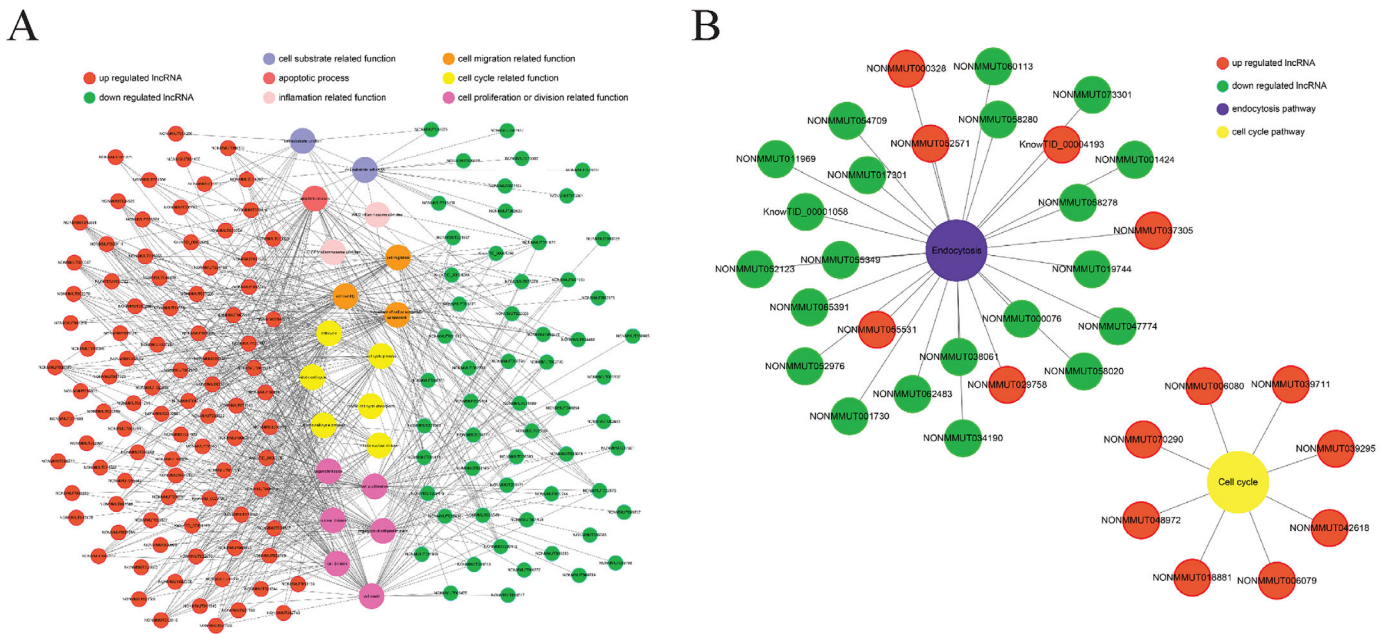


Figure 2. Enrichment analysis of the top 200 dysregulated lncRNA candidates, both upregulated and downregulated, using GO and pathway analysis. **(A)** Construction of a network based on selected lncRNAs associated with several GO terms. The selection was made using GO analysis results related to the *cis*-associated genes of these lncRNAs. **(B)** Construction of a network based on selected lncRNAs associated with several KEGG pathway terms. The selection was made using pathway analysis results related to the *cis*-associated genes of these lncRNAs.

GO biological processes primarily involved cell cycle, cell migration, cell proliferation, and assembly of cellular components. The corresponding lncRNAs were connected with their respective GO terms on the enrichment network (Fig. 2A). Furthermore, the predicted *cis*-associated genes underwent pathway enrichment analysis based on the KEGG database, aiming to elucidate complex regulatory networks. Pathways associated with signal transduction and diseases that showed significant enrichment compared to the background were identified using a threshold of $P \leq 0.05$. The KEGG pathway terms of interest were then linked with their corresponding lncRNAs to construct the enrichment network (Fig. 2B).

Lentiviral-Mediated Knockdown of linc17500 in Mouse Corneal Epithelial Cells

To investigate the functional role of linc17500 in mouse corneal epithelial cells, we successfully generated recombinant lentiviruses carrying sh-linc17500-1, sh-linc17500-2, and a negative control (sh-NC). These lentiviruses contained the green fluorescent protein (GFP) gene, allowing for the sorting of TKE2 cells infected with Lv-sh-linc17500 and Lv-sh-NC (multi-

plicity of infection = 50) via flow cytometry, specifically targeting cells exhibiting high fluorescence signals (Fig. 3A). Subsequent RT-qPCR analysis demonstrated that the expression levels of linc17500 in the two Lv-sh-linc17500-infected TKE2 cell groups were 27.5% and 9.0% of the expression level observed in Lv-sh-NC-infected TKE2 cells (Fig. 3B). These findings validate the successful establishment of TKE2 cells displaying significantly and consistently reduced levels of linc17500 expression, along with their respective control counterparts.

Knockdown of linc17500 Inhibits the Proliferation and Migration of TKE2 Cells

Given that cell proliferation and migration are critical factors affecting the rate of wound healing, we conducted MTS and EdU assays to evaluate the impact of linc17500 depletion on TKE2 cell proliferation. The results demonstrated a significant suppression of cell proliferation in TKE2 cells upon linc17500 inhibition (Figs. 4A, 4D, 4E). Furthermore, in line with the MTS and EdU assay findings, flow cytometry analysis revealed that the percentages of Lv-sh-linc17500-infected TKE2 cells in the G1/S phase were 69.3% and 80.9%, whereas the control group cells exhibited a value

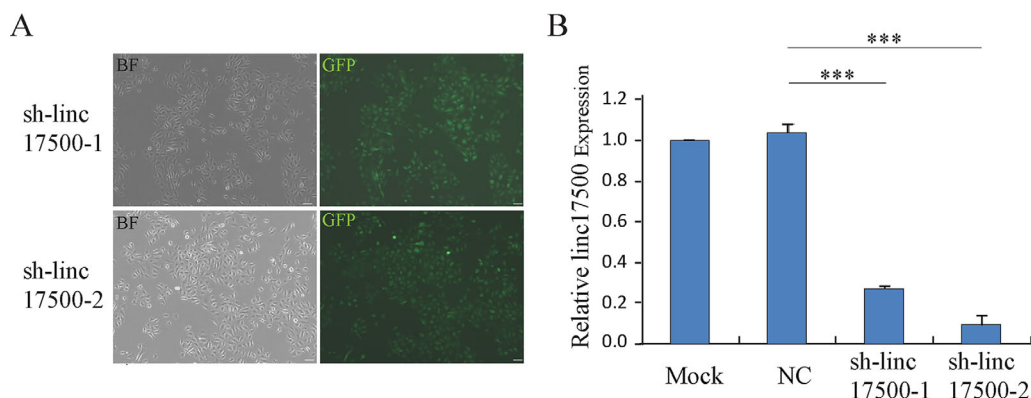


Figure 3. Knockdown of linc17500 using lentiviruses carrying shRNA targeting linc17500. **(A)** Assessment of viral transfection efficiency based on GFP expression. **(B)** Measurement of linc17500 expression levels in TKE2 cells infected with Lv-sh-linc17500 and Lv-sh-NC using RT-qPCR ($n = 3/\text{group}$). Scale bars: 50 μm .

of 54.1% (Figs. 4B, 4C). These observations suggest that linc17500 inhibition induced cell-cycle arrest at the G1/S phase in TKE2 cells. To further investigate the effect of linc17500 inhibition on cell migration, we conducted a scratch-wound assay across different

groups. After 24 hours, the remaining unhealed areas in the Lv-sh-linc17500-infected TKE2 cells were larger compared to those in the control group, indicating a significant reduction in cell migration upon linc17500 inhibition (Figs. 4F, 4G).

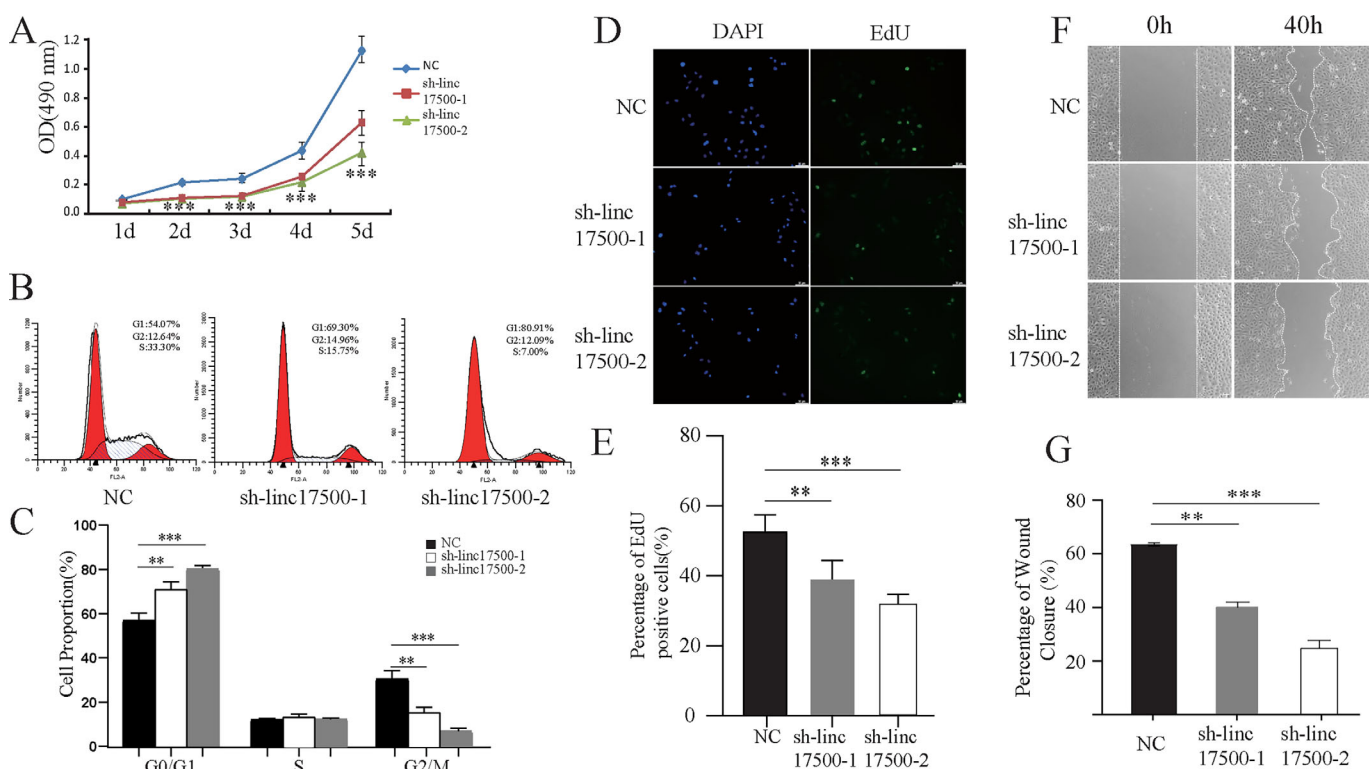


Figure 4. Effects of linc17500 knockdown on TKE2 cell proliferation and migration in vitro. **(A)** Assessment of the effects of linc17500 knockdown on TKE2 cell proliferation using an MTS assay. **(B, C)** Evaluation of the effects of linc17500 downregulation on the distribution of TKE2 cell-cycle phases using flow cytometry. **(D)** Immunofluorescence images display DAPI (blue) staining for nuclei and EdU (green) incorporation indicating proliferation in TKE2 cells infected with Lv-sh-NC or Lv-sh-linc17500. **(E)** Statistical analysis of proliferating cells in Lv-sh-NC or Lv-sh-linc17500 infected groups ($n = 6/\text{group}$). **(F)** Determination of the effects of linc17500 knockdown on TKE2 cell migration using a scratch-wound assay. **(G)** Quantification of the percentage of wound closure (%) by normalizing the healed area to the initial wound area ($n = 3/\text{group}$). Scale bars: 50 μm .

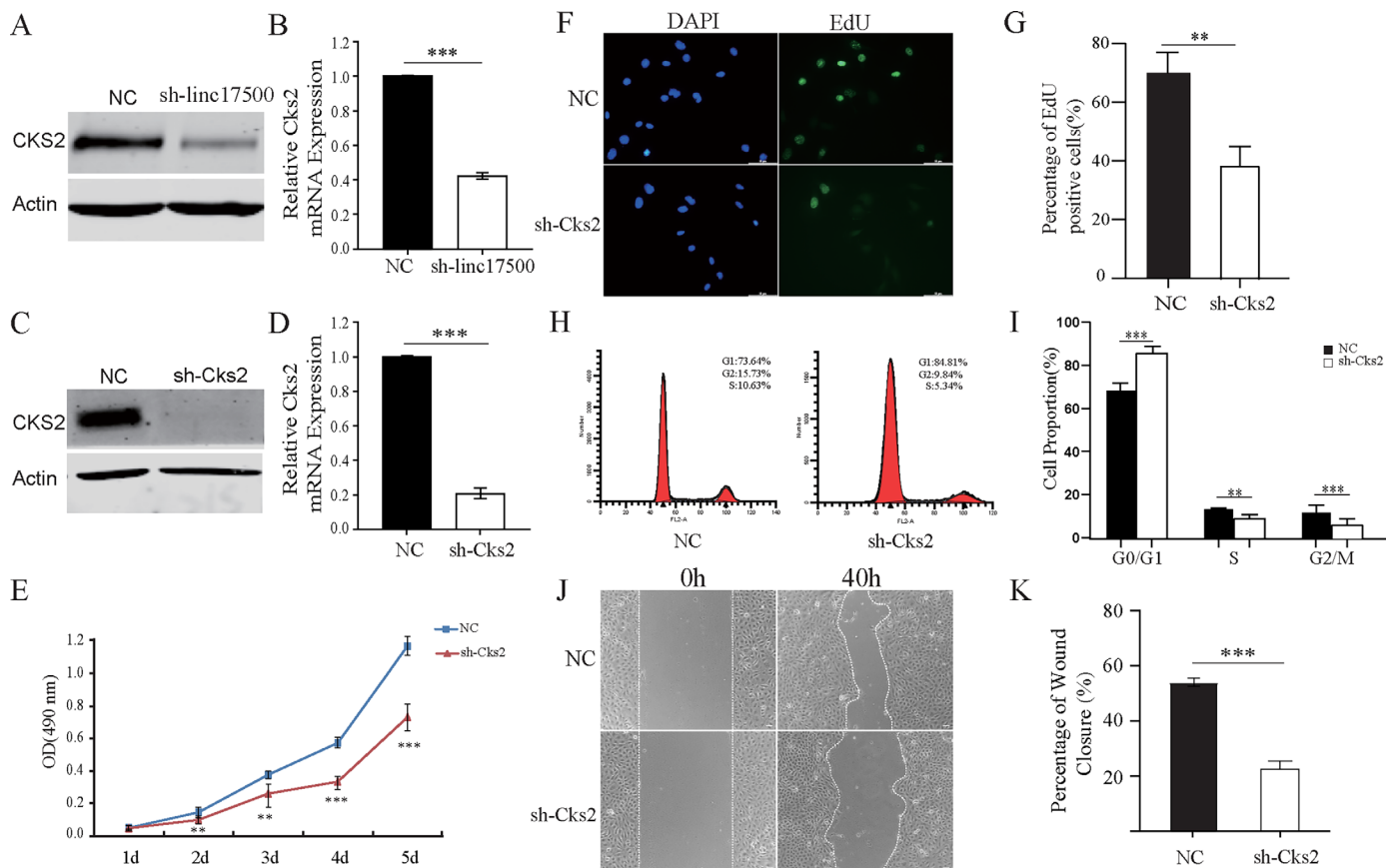


Figure 5. Identification of *Cks2* as a *cis*-associated gene of *linc17500* and its effects on TKE2 cell proliferation and migration. (A, B) RT-qPCR and WB analyses were performed to quantify *Cks2* mRNA and protein levels in TKE2 cells infected with either Lv-sh-NC or Lv-sh-*linc17500*. (C, D) RT-qPCR and WB assays measured *CKS2* expression post-infection with lentiviral vectors harboring shRNA against a non-targeting control (Lv-sh-NC) and *Cks2* (Lv-sh-*Cks2*). (E) MTS assays assessed TKE2 cell proliferation following *CKS2* knockdown. (F) Representative immunofluorescence images showing DAPI-stained nuclei (blue) and EdU incorporation (green) in TKE2 cells infected with Lv-sh-NC and Lv-sh-*Cks2*, indicative of active DNA synthesis. (G) Statistical comparison of the percentage of EdU-positive cells, indicating proliferation rates between the two groups ($n = 6$ /group). (H, I) Flow cytometric analysis evaluated TKE2 cell-cycle distribution changes subsequent to *CKS2* suppression. (J) Scratch-wound assays determined the migratory capacity of TKE2 cells after *CKS2* silencing. (K) Quantitative analysis of wound healing, expressed as the percentage of wound closure, normalized to the initial wound area ($n = 3$ /group). Scale bars: 50 μ m.

Characterizing *Cks2* as a *cis*-Associated Gene of *linc17500* and Its Role in TKE2 Cell Behavior

Given the potential role of *linc17500* in modulating TKE2 cell proliferation and migration through its *cis*-associated genes, we identified *CKS2* as a pertinent target based on the enrichment analysis. Importantly, the silencing of *linc17500* led to significant alterations in both *Cks2* mRNA and protein expression levels (Figs. 5A, 5B), suggesting a transcriptional regulatory mechanism by which *linc17500* controls *CKS2*. To elucidate the biological function of *CKS2*, we generated TKE2 cell lines with stable knockdown of *CKS2*. These cells exhibited a marked reduction in *CKS2* protein expression (Figs. 5C, 5D). Proliferative capacity, assessed via MTS and EdU incorporation assays,

was significantly diminished in *CKS2*-depleted cells (Figs. 5E–5G). Additionally, flow cytometric analysis indicated that *CKS2* knockdown resulted in a pronounced G1/S phase cell-cycle arrest (Figs. 5H, 5I). In parallel, wound healing assays demonstrated that *CKS2* silencing led to a substantial decrease in TKE2 cell migratory ability (Figs. 5J, 5K). These findings collectively suggest that *CKS2* is a critical effector in the *linc17500* regulatory axis, mediating essential processes of cell-cycle progression and cellular motility in CEWH.

Downregulation of *linc17500* Leads to an Inactivation of Intracellular Rb and Akt

To investigate the impact of *linc17500* depletion on intracellular signal pathways, we assessed

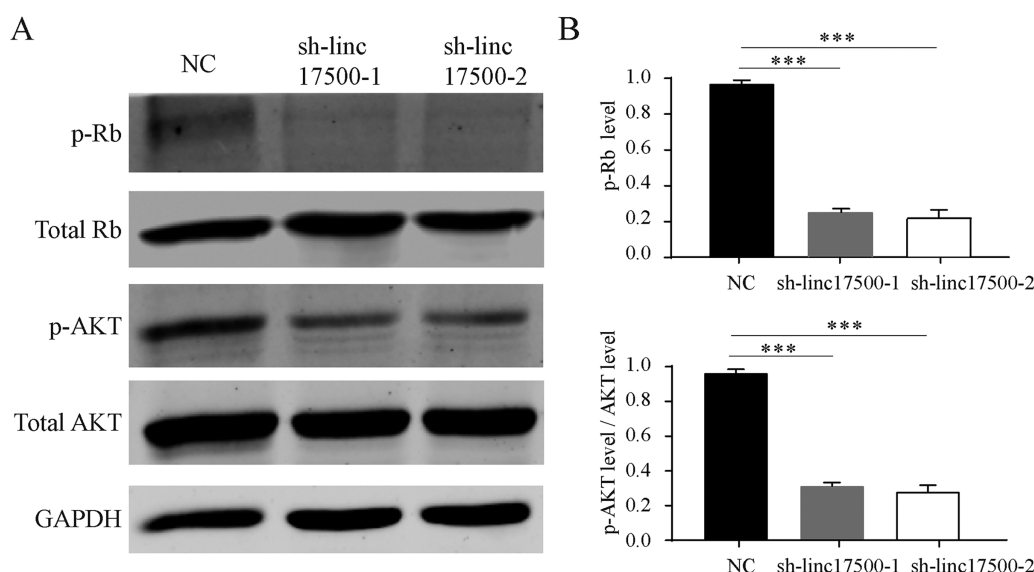


Figure 6. Suppression of Rb and Akt activation status following linc17500 knockdown in TKE2 cells. **(A)** WB analysis depicting the protein expression levels of p-Rb, Rb, p-Akt, Akt, and GAPDH. **(B)** Densitometric analysis of WB results quantifying the relative levels of p-Rb, Rb, p-Akt, and Akt ($n = 3/\text{group}$).

the activation status of Rb and Akt in TKE2 cells. Cell extracts were subjected to WB analysis using phosphorylation-specific antibodies against Rb and Akt, key molecules involved in cell proliferation and migration. As depicted in Figures 6A and 6B, the knockdown of linc17500 resulted in decreased expression levels of phosphorylated Rb and Akt in TKE2 cells, while total Rb and Akt protein levels remained unchanged. These findings collectively indicate that linc17500 depletion in TKE2 cells suppresses the activation status of Rb and Akt, thereby inhibiting cell proliferation and migration.

Knockdown of linc17500 Delays Corneal Epithelial Wound Healing In Vivo

To confirm the essential role of linc17500 upregulation in CEWH, we conducted intrastromal injections of si-linc17500 and si-NC, establishing in vivo CEWH models following established protocols. After 24 hours, the average percentage of healed area was 37.8% in the si-linc17500 injection group, significantly lower than the 59.2% observed in the si-NC group (Figs. 7A, 7B). This disparity clearly indicates that linc17500 inhibition decelerates the process of corneal epithelial wound healing. To further validate this observation, we scraped the regenerated corneal epithelia and assessed linc17500 expression using RT-qPCR. The results demonstrated reduced linc17500

expression in the corneas injected with si-linc17500 compared to the contralateral corneas injected with si-NC (Fig. 7C). H&E and immunostaining utilizing Krt12, Krt14, and Ki67 were conducted to evaluate re-epithelialization, proliferation, and migration within the corneal epithelium during CEWH. H&E staining revealed that, compared to the intact corneal structure comprised of layers of basal, wing, and superficial cells, the cornea exhibited a reduced epithelial thickness of only one or two cell layers at 24 hours post-injury. Notably, the epithelial coverage in corneas treated with si-linc17500 was significantly diminished relative to those treated with si-NC (Supplementary Fig. S1A). In the intact cornea, Krt12 was ubiquitously expressed across all corneal layers, whereas Krt14 expression was confined to the basal epithelial cells (Supplementary Fig. S1B). Following injury, both Krt12 and Krt14 were prominently expressed in the epithelial cells bridging the wound area at 24 hours post-injury (Supplementary Fig. S1B). Ki67 immunostaining demonstrated no significant difference in the proliferation rate, as indicated by the number of Ki67-positive cells, between corneas treated with si-NC and those treated with si-linc17500 (Supplementary Fig. S1B). The consistent effects of linc17500 on wound healing observed in both in vitro and in vivo experiments provide strong evidence supporting its critical role in promoting cell proliferation and migration during CEWH.

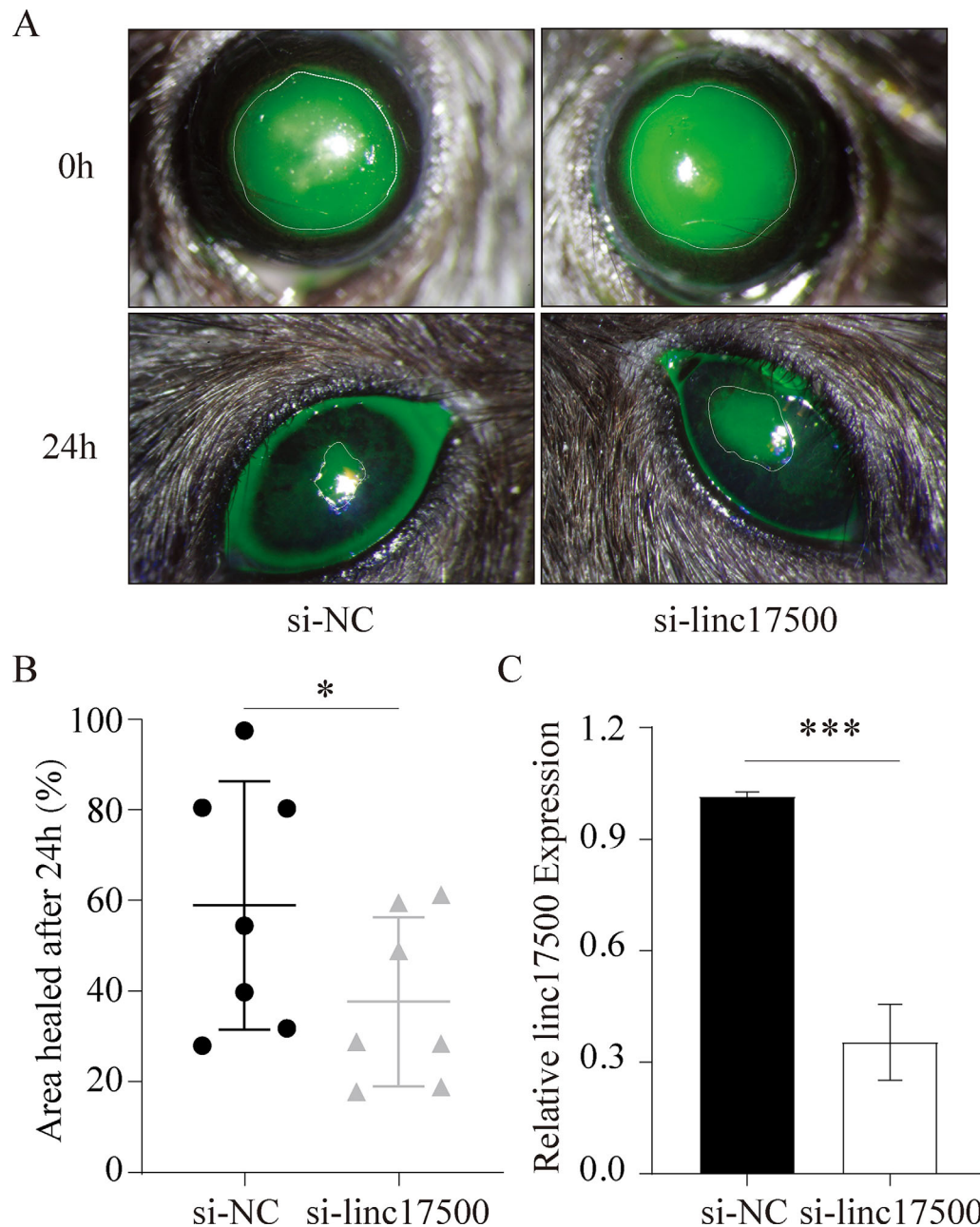


Figure 7. Delayed corneal epithelial wound healing following linc17500 knockdown. **(A)** Representative images of sodium fluorescein staining depicting the remaining unhealed area 24 hours after injury. The wound edges are demarcated in white. **(B)** Quantification of the percentage of wound closure (%) by normalizing the healed area to the initial wound area ($n = 7/\text{group}$). **(C)** RT-qPCR analysis evaluating the expression level of linc17500 in each murine corneal epithelium ($n = 7/\text{group}$).

Discussion

Despite the continual expansion of our understanding of epigenetic regulatory mechanisms in CEWH, there remains a dearth of systematic analyses concerning lncRNA involvement in this process. In this study, we conducted the first comprehensive profiling of lncRNA expression during mouse CEWH using a

microarray assay. The identified lncRNA candidates, predominantly upregulated during CEWH, underwent rigorous GO and KEGG pathway enrichment analyses, revealing their active participation in various crucial regulatory responses associated with CEWH. Remarkably, our findings underscore the indispensable role of lncRNAs in CEWH, as demonstrated by the delayed corneal epithelial wound healing observed both *in vivo* and *in vitro* upon knockdown of the linc17500.

In general, lncRNAs exert their regulatory effects through either *cis*-acting regulation of neighboring protein-coding genes or *trans*-acting regulation of distant protein-coding genes.^{23,24} In our study, we successfully established the lncRNA expression profiles and performed *cis*-associated gene enrichment analysis for the top 200 upregulated and top 200 downregulated lncRNAs. Notably, these lncRNAs exhibited significant enrichment for functions associated with wound healing, a process that prominently relies on corneal epithelial cell proliferation, migration, adhesion, and differentiation.^{1,25} Of particular interest, CDC28 protein kinase regulatory subunit 2 (CKS2), a protein frequently overexpressed in a variety of malignancies and known to contribute to the proliferation and migration of numerous cancer cell types,²⁶ was identified as a *cis*-associated gene of lncRNA NONMMUT017500. The regulatory influence of CKS2 on CEWH is substantiated by our findings that functional attenuation of CKS2 markedly reduces TKE2 cell proliferation and migration in vitro. Intriguingly, our data suggest that linc17500 may exert regulatory effects on CEWH through CKS2-dependent pathways. Similarly, bone morphogenetic protein 3 (BMP3), which plays a critical role in epidermal cell differentiation and development,^{27,28} was found to be a *cis*-associated gene of lncRNA NONMMUT053228. Remarkably, our enrichment analysis revealed that these lncRNAs and their *cis*-associated genes displayed higher expression levels in the repaired injured corneas compared to the contralateral corneas. These associations strongly suggest the involvement of these lncRNAs in the wound healing process, potentially regulating the proliferation and differentiation of corneal epithelial cells. Furthermore, considering that the mechanisms underlying wound healing may exhibit shared characteristics across different organs or tissues as a normal physiological response to tissue damage, our lncRNA expression profiling in CEWH not only expands our understanding of the epigenetic mechanisms underlying CEWH but also holds potential for providing novel insights into other wound disorders.

Recent investigations have shed light on the emerging role of lncRNAs in skin wound healing. Notably, lncRNA metastasis-associated lung adenocarcinoma transcript 1 (MALAT1) and lncRNA H19 have been found to accelerate the healing process in skin wounds of diabetic mice, operating through the hypoxia-inducible factor-1 alpha (HIF-1 α) signaling pathway.^{29,30} Another study revealed the involvement of lncRNA wound and keratinocyte migration-associated lncRNA 1 (WAKMAR1) in regulating human skin wound healing by enhancing keratinocyte

migration through interference with the promoter methylation of the *E2F1* gene.³¹ Additionally, lncRNA XIST was found to promote extracellular matrix synthesis, proliferation, and migration in human skin fibroblasts following thermal injury by acting as a miR-29b-3p sponge and targeting COL1A1.³² Moreover, a study by Chen and Hu³³ investigated the expression profile of differentially expressed lncRNAs and revealed that lncRNA 3632454L22RiK may positively impact diabetic corneal epithelial wound healing by sponging miR-181a-5p. Collectively, these findings underscore the intriguing and significant potential of lncRNAs to serve as a promising avenue for future investigations into wound healing processes.

Among the diverse array of lncRNAs, linc17500 emerges as one of the most abundant lncRNAs in corneal epithelial cells and exhibits the most prominent upregulation during CEWH. Subsequent target gene enrichment analysis of linc17500 suggests its potential involvement in critical cellular processes such as cell-cycle progression, cell proliferation, and cell migration. These findings provide valuable insights into the functional implications of linc17500 in CEWH. Notably, our data indicate that depletion of linc17500 in TKE2 cells hinders G1/S phase transition and suppresses cell proliferation, resulting in decreased levels of phosphorylated Rb. Rb, a pivotal tumor suppressor factor governing progression through the G1/S phases, plays a critical role in cell-cycle regulation. A recent study demonstrated that growth arrest-associated lncRNA 1 (GASL1), a novel lncRNA, inhibits cell proliferation and restricts E2F1 activity by reducing Rb phosphorylation.³⁴ Additionally, the H19-derived miR-675 has been implicated in colorectal cancer through downregulation of its target Rb.³⁵ Furthermore, depletion of linc17500 impedes cell migration in TKE2 cells and leads to reduced levels of phosphorylated Akt. Activation of the phosphoinositide 3-kinase (PI3K)/Akt signaling pathway, characterized by elevated expression of p-Akt, holds crucial significance in cell migration and exerts notable effects on cutaneous wound healing processes, including regeneration³⁶ and remodeling.³⁷ Notably, a study by Zhang et al.³⁸ revealed that lncRNA long intergenic non-protein coding RNA 672 (LINC00672) facilitates the conversion of human skin fibroblasts into keratinocyte-like cells through the PI3K/Akt signaling pathway. Moreover, the involvement of lncRNAs in the PI3K/Akt signaling pathway has been documented in various diseases. For example, the lncRNA maternally expressed gene 3 (MEG3) has been shown to alleviate neuronal damage and improve cognitive impairment in hippocampal tissues of Alzheimer's disease rats by inactivating the PI3K/Akt signaling pathway.³⁹

The mechanisms employed by lncRNAs to regulate gene expression have garnered considerable attention, involving direct interactions with RNA molecules, proteins, and chromatin. For example, the lncRNA HOX antisense intergenic RNA (HOTAIR) recruits the polycomb repressive complex 2 (PRC2) to the HoxD gene cluster, thereby modulating chromosome occupancy through enhancer of zeste homolog 2 (EZH2).⁴⁰ Lopez-Pajares et al.⁴¹ demonstrated that two lncRNAs—namely, anti-differentiation non-coding RNA and terminal differentiation-induced non-coding RNA—exert regulatory effects on key transcription factors musculoaponeurotic fibrosarcoma (MAF) and MAFB upstream of epidermal differentiation, implicating their involvement in skin wound healing. Additionally, the lncRNA myocardial infarct-associated transcript (MIAT) has been found to be upregulated in the diabetic retina and acts as a competing endogenous RNA (ceRNA) by sequestering miR-150-5p.⁴²

It is worth noting that this study has also uncovered several lncRNAs with alterations as notable as those of linc17500 during CEWH, such as NONMMUT68561, NONMMUT34186, and NONMMUT34190, all of which merit further detailed investigation. However, these candidates posed significant technical challenges for genetic overexpression and knockdown. Recognizing these hurdles, we are committed to addressing the roles of these lncRNAs in future studies as we develop and refine methodologies to overcome these limitations.

Conclusions

This study emphasizes the significance of lncRNAs, particularly linc17500, in the context of CEWH, thereby offering valuable insights for future investigations into the epigenetic mechanisms underlying CEWH. Moreover, our findings have the potential to identify a novel class of biomarkers for assessing CEWH or serve as potential therapeutic targets for the clinical treatment of corneal wound complications.

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